QUALITY ENGINEERING · DATA INTEGRATION

Banking Data Warehouse ETL Test Plan

Test activities, scenario coverage, execution controls, and review criteria

DRAFT — PENDING HUMAN REVIEW

Project	Banking ETL Testing
System under test	Banking ETL flow to SQL Server Data Warehouse
Document version	1.0 — Initial draft
Prepared by	QA / Test Planning (owner to confirm)
Prepared date	02-Nov-2026
Primary requirements	Banking Data Warehouse BRD v1.0
Strategy basis	Banking ETL Test Strategy v1.0 — Draft, pending human review

This plan was prepared at the user's direction using the available draft strategy. It is not approval of that strategy, a commitment to dates, or evidence that testing has been executed.

Test Plan

Purpose: Plan the verification of the Banking ETL flow that integrates customer, account, transaction, and branch data into a centralized SQL Server Data Warehouse. This plan defines proposed phases, coverage, readiness criteria, and reporting; it is a planning document, not a set of detailed test cases.

Traceability convention: BR-01 through BR-10 below correspond to the numbered high-level business requirements in the BRD. Test scenario identifiers TP-01 through TP-12 map to those requirements or to the BRD's stated source-file and ETL testing scope.

> **Governance:** The source test strategy is explicitly marked Draft — Pending Human Review. This plan is therefore also a draft and must not be baselined as approved until the strategy and the open requirements in this plan have been reviewed by authorized owners.

Objective

Verify, using approved mappings and business rules, that the ETL process loads accurate, complete, and consistent banking data, enforces the defined relationships, handles invalid/duplicate records as specified, processes incremental changes correctly, and records execution failures and rejected records.

- Validate the four named source files and their CSV/Excel structures.
- Confirm valid customers, accounts, transactions, and branch reference data reach their intended warehouse tables.
- Test that account/customer and transaction/account relationships satisfy approved rules.
- Reconcile source, accepted, rejected, and target populations; verify transaction amount accuracy.
- Exercise incremental processing, failure handling, logging, recovery, and repeat-run behavior once technical rules are supplied.

Scope

In scope

- Source file receipt/readability, CSV and Excel structure, source profiling, and malformed/missing-file handling.
- Approved source-to-target mapping, transformations, data types, precision, key relationships, and data-quality rules.
- Full and incremental ETL runs; completeness, accuracy, integrity, reconciliation, and regression coverage.
- Invalid/duplicate data disposition, reject/error-table behavior, package failure logging, and restart/replay behavior.
- Test evidence, defect reporting, retesting, and test closure reporting.

Out of scope unless the owners add them

- Testing of upstream banking applications and downstream reports, dashboards, or user interfaces.
- Formal performance/volume, penetration/security, disaster-recovery, availability, accessibility, and localization test phases; applicable requirements and targets have not been supplied.
- Business calculations, financial interpretations, or transformation outcomes not stated in approved requirements or mapping documents.

> **Not assumed:** The template's Restful Booker API scope, browser/device matrix, API CRUD operations, named environments, named staff, Jira use, and other sample-project details do not describe this banking ETL project and are not carried forward as project facts.

Inclusions

Plan ID	Proposed scenario / activity (not a detailed test case)	Trace	Priority	Readiness / expected-result dependency
TP-01	Validate delivery, readability, and approved schema/header/layout for customers.csv, accounts.csv, transactions.csv, and branch_master.xlsx; check agreed empty, missing, and malformed input handling.	BRD §2; ETL scope	TBD	File contracts and delivery rules required.
TP-02	Verify valid customer records load to Dim_Customer and reconcile source/accepted/target counts.	BR-01, BR-08	TBD	Customer validity, keys, mapping, and reject policy required.
TP-03	Verify each loaded account belongs to a valid customer; cover valid, missing, and invalid customer references.	BR-02, BR-06	TBD	Business key and invalid-reference disposition required.
TP-04	Verify each transaction references a valid account; cover valid, orphan, and boundary/reference cases allowed by approved rules.	BR-03, BR-06	TBD	Account matching, transaction rules, and reject policy required.
TP-05	Validate branch data against the branch master, including the agreed matching key and invalid/unmatched outcomes.	BR-04	TBD	Branch key, matching semantics, and reject treatment required.
TP-06	Exercise duplicate source records and verify the specified deduplication, rejection, retention, and audit behavior.	BR-05	TBD	Duplicate key and precedence/disposition are not defined.
TP-07	Exercise invalid records and verify routing to the approved reject/error destination, including countability and error detail.	BR-06, BR-10	TBD	Invalidity rules, destination, partial-load policy, and error schema required.
TP-08	Compare transaction amounts from source to Fact_Transaction, including approved conversion, precision, scale, and rounding cases.	BR-07	TBD	Currency, precision, sign, conversion, and rounding rules required.
TP-09	Reconcile source and target record counts by entity and agreed batch, accounting for accepted, rejected, and duplicate records.	BR-08	TBD	Comparison grain, accounting formula, and tolerance required.
TP-10	Verify incremental insert/change processing, unchanged reruns, repeat execution, and any agreed late-arriving or correction behavior.	BR-09	TBD	Change detection, watermark, replay, and history rules required.
TP-11	Induce an approved recoverable package failure and verify visible run status, failure/reject logging, recovery, and safe restart without unintended duplication or partial data.	BR-10; ETL scope	TBD	Failure points, retry/restart model, and log requirements required.
TP-12	Rerun impacted file, transformation, relationship, reconciliation, and incremental coverage after an ETL change or defect fix.	BRD §5; strategy basis	TBD	Approved build baseline and impact assessment required.

Priority: P0/P1/P2 values remain TBD until business and QA owners approve impact/urgency criteria. No priority has been inferred from the example template.

Test Environments

Environment / asset	Known information	Details required before execution
Source files	customers.csv, accounts.csv, transactions.csv, branch_master.xlsx	Approved layouts, sample/controlled test fixtures, delivery path, versioning, and arrival behavior.
ETL runtime	Not identified in the BRD.	Tool/version, package or pipeline build, parameters, execution method, and relevant dependencies.

Environment / asset	Known information	Details required before execution
Target warehouse	SQL Server; expected tables Dim_Customer, Dim_Account, Dim_Branch, Fact_Transaction	Test endpoint/version, database/schema, table definitions, access, refresh/reset method, and connectivity.
Execution / logging	Package failures and rejected records must be logged.	Run environment, log location/schema, monitoring access, evidence retention, and failure/restart controls.
Test data	No sample files or target extracts were present in the supplied project documents.	Approved representative, positive, negative, boundary, duplicate, orphan, and incremental datasets; expected results and privacy controls.

Use masked or synthetic banking/customer data unless the data owner expressly authorizes another approach. Apply approved least-privilege access and avoid exposing sensitive values in defect evidence. Environment names, URLs, OS/browser matrix, and test dates are TBD; API-template examples are not project configuration.

Defect Reporting Procedure

1. Log a defect when an observed result deviates from an approved requirement, mapping, business rule, expected result, or agreed operational behavior.
2. Record a unique defect ID, linked BR/TP reference, ETL build/environment, batch/run and source-file identifiers, reproducible steps, sanitized test-data reference, expected and actual results, impact/severity, priority, evidence/log reference, owner, and status.
3. Triage with QA, ETL, data, and business owners; agree severity, priority, fix target, and retest/regression scope. Severity and triage definitions are TBD.
4. After a fix, retest the failure, reconcile impacted data, and run regression coverage selected by impact assessment. Record results and close only with evidence.
5. Publish a status summary at an agreed cadence; tracker, workflow, communication channel, escalation path, and reporting cadence are TBD.

> **Tooling:** The example mentions Jira, but no project defect tracker was confirmed. Use the approved project tracker once the owner identifies it; do not assume Jira is provisioned.

Test Strategy

Execution is requirements-based and risk-informed, following the provided Banking ETL Test Strategy draft as a provisional planning reference. That strategy is not approved; its scope, roles, and proposed controls require human review. BRD requirement coverage is summarized below.

BRD trace	Requirement summary	Planned coverage	Open dependency
BR-01	Load all valid customers.	TP-02; valid/invalid customer handling and count reconciliation.	Validity rules, mapping, keys.
BR-02	Each account belongs to a valid customer.	TP-03; referential integrity and invalid-reference disposition.	Relationship key and reject rules.
BR-03	Each transaction references a valid account.	TP-04; referential integrity and orphan handling.	Relationship key and reject rules.
BR-04	Branch information is validated against branch master.	TP-05; branch match and unmatched outcomes.	Matching semantics and keys.

BRD trace	Requirement summary	Planned coverage	Open dependency
BR-05	Handle duplicates according to business rules.	TP-06; repeatable duplicate outcome.	Duplicate policy is explicitly pending.
BR-06	Reject invalid records or move them to an error table.	TP-03, TP-04, TP-07; reject/error accounting.	Destination, criteria, batch atomicity.
BR-07	Load transaction amounts accurately.	TP-08; field-level and aggregate validation.	Precision, currency, rounding.
BR-08	Reconcile source and target counts.	TP-02, TP-09; entity/batch count reconciliation.	Reconciliation formula and tolerance.
BR-09	Process incremental changes correctly.	TP-10; insert/change/replay scenarios.	Watermark and change/history rules.
BR-10	Log package failures and rejected records.	TP-07, TP-11; failure, reject, and recovery evidence.	Required log fields and restart model.

Use source/file checks, smoke validation of a test build, mapping/data-quality checks, package and integration execution, relationship checks, reconciliation, negative testing, incremental/recovery testing, retesting, and impact-based regression. Apply boundary analysis, equivalence partitioning, decision tables, and exploratory testing where supported by approved rules. Do not invent expected values or acceptance thresholds.

Test Schedule

Milestone / activity	Planned order / dependency	Target dates / duration	Exit evidence
Review plan, draft strategy, BRD, and open requirements	First; obtain owner decisions before baseline.	TBD by project owner.	Reviewed plan; strategy approval status recorded; questions assigned.
Finalize STTM, business rules, file contracts, and acceptance criteria	Before expected outcomes and detailed case design are baselined.	TBD; dependency-driven.	Versioned, owner-approved specifications.
Provision environment, access, test files, and target baseline	Before test execution.	TBD; dependency-driven.	Readiness confirmation and controlled data/evidence.
Prepare and review scenarios / detailed cases	After requirement baseline; case design is a subsequent activity, not included here.	TBD by QA lead.	Traceability and expected-result review.
Execute test cycles, triage defects, retest, and regress	After entry criteria pass; repeat cycles as needed.	TBD by release/sprint plan.	Execution results, defects, reconciliation and regression evidence.
Publish test summary and close	After exit criteria review and approver decision.	TBD by project owner.	Summary, residual risks, exceptions, and recorded sign-off.

No sprint count, duration, start/end date, or resource allocation is confirmed in the BRD; the template's sample schedule is not applied.

Test Deliverables

- This Test Plan, pending review and approval.
- After plan/strategy approval: traceable scenarios and detailed test cases with requirement links and approved expected outcomes.
- Approved test-data inventory and environment/readiness record; sanitized source/target reconciliation evidence.

- Execution results, package/run logs as authorized, defect records, retest/regression results, and blocked/not-run rationale.
- Test execution summary, residual risk/exception register, and human approval record.

Detailed test cases, SQL validation scripts, and automation are deliberately not included in this planning document.

Entry and Exit Criteria

Requirement analysis / planning

Entry: BRD and the available strategy/template are identified. **Exit:** Open requirements, strategy approval status, scope, owners, and blocking dependencies are documented; owners decide whether the plan can be baselined.

Test design and readiness

Entry: Approved strategy and in-scope requirements are baselined. **Exit:** STTM, business rules, file contracts, expected outcomes, scenario traceability, test data, environment, access, logging, defect workflow, and run/reset process are available and reviewed.

Test execution

Entry: Test build is identified; readiness checks pass; reviewed cases and controlled data are available. **Exit:** Planned tests are executed or explicitly classified as blocked/not run; evidence, defect states, retests, regression, and approved reconciliations are recorded.

Test closure

Entry: Execution and defect/reconciliation reports are ready. **Exit:** Authorized owners review summary, unresolved defects, exceptions, residual risks, and business impact; closure/sign-off is recorded. No exit percentage or severity threshold is set until owners approve one.

Test Execution

1. Confirm the approved build, test environment, file contract, source data, target baseline, credentials/access, and run-log availability.
2. Run a controlled smoke load and check completion status, basic target population, and logging. Stop and report if a critical readiness or integrity failure occurs.
3. Execute source structure, mapping, data-quality, reference-integrity, amount, and reject scenarios against approved expected outcomes.
4. Run full-load reconciliation, then incremental, replay, failure/restart, and regression coverage according to the agreed dependency order.
5. Record pass/fail/blocked/not-run status, evidence, requirement trace, batch/run IDs, defects, and reconciliation results. Retest fixes and assess regression impact.
6. Report execution progress and blockers at the cadence agreed by the project; do not claim a scenario passed without objective evidence.

> **Stop / hold condition:** Hold affected testing if strategy approval is outstanding, mapping or expected results are ambiguous, test data is unauthorized, or environment/logging cannot support safe and traceable execution. Escalate to the named owner rather than inventing a rule.

Test Closure

Prepare a closure summary showing planned versus executed scenarios, pass/fail/blocked/not-run counts, coverage by BR-01–BR-10, reconciliation outcomes, defects by agreed severity/status, retest and regression results, environment/build, evidence location, known limitations, risk acceptances, and required approvals. Close only after authorized reviewers accept remaining risks and record the decision. Closure does not imply the warehouse or strategy is formally approved.

Tools

Capability	Known / proposed use	Project decision
SQL Server	Confirmed target data platform; database validation will use the project-approved access method.	Instance/version, schemas, client, access roles, and query/evidence controls TBD.
ETL execution / scheduling	Required to run and observe package/pipeline behavior.	Product, versions, scheduler, and logs not identified.
Defect / project tracking	Needed to record defects, owners, and approvals.	Tracker and workflow not identified; Jira in the example is not confirmed.
Test documentation / evidence	Use approved project-controlled storage for scenarios, results, and sanitized evidence.	Repository, retention, and access policy TBD.

Risks and Mitigations

Risk / dependency	Impact	Mitigation / action
Test Strategy remains unapproved.	Plan may not align with authorized scope, approach, or exit controls.	Obtain QA/business/technical review; baseline plan only after strategy and scope are confirmed.
STTM, target schema, and detailed business rules are absent.	Transformation, field-level, and relationship expected results cannot be verified.	Obtain owner-approved mapping/schema/rules before test design and execution.
Duplicate and reject rules are incomplete.	Accepted/rejected outcomes and reconciliation can be disputed.	Agree duplicate keys/precedence, invalidity criteria, reject destination, logging, and partial-load behavior.
Incremental and recovery semantics are unspecified.	Missed/duplicated changes, unsafe replay, and incomplete recovery may escape detection.	Document change keys, watermark, history, late-arrival, retry, and restart behavior; test against an approved baseline.
Environment, source fixtures, and access are unavailable or late.	Execution may be blocked or non-representative.	Assign provision/data owners; prepare masked or synthetic representative data and environment readiness checks.
Transaction amounts or sensitive data are mishandled.	Financial data accuracy or confidentiality may be compromised.	Approve amount precision/currency rules; restrict access, mask data, and sanitize evidence.
Schedule, staffing, severity thresholds, and approvals are unknown.	Resourcing and release/closure decisions cannot be planned.	Project owner to provide schedule, assigned roles, triage workflow, and measurable exit/sign-off criteria.
Performance/security/non-functional requirements are unstated.	Relevant risks may remain untested or be incorrectly treated as in scope.	Business/operations/security owners decide applicable coverage and targets before plan approval.

Approvals

Review and approval of the Test Strategy is a prerequisite to baselining this Test Plan. Test design, execution, and closure proceed only after the relevant phase approvals and readiness criteria are met.

Approver role	Name	Decision / comments	Date
QA / Test Lead	TBD	Pending review	TBD
Business Owner / Data Owner	TBD	Pending review	TBD
ETL / Technical Owner	TBD	Pending review	TBD

Gaps and Questions for the Author

#	Missing / ambiguous information and impact	Question for owner	Blocks
1	The only available Test Strategy is a draft pending review; this plan cannot claim alignment to an approved baseline.	Who will approve or revise the strategy, and what scope/criteria should this plan inherit?	Plan baselining.
2	No STTM, target table schema, or complete mapping was present.	Can the data/ETL owner provide approved columns, keys, transformations, defaults, and target constraints?	Detailed expected results and transformation execution.
3	Validity, referential, branch-matching, and edge-case rules are high-level only.	What fields, valid domains, relationship keys, branch match rules, and boundary/empty conditions define acceptance?	Data quality and referential pass/fail.
4	Duplicate and invalid-record treatment is not defined.	What are the deduplication key/precedence, reject criteria/destination, error schema, and partial-load policy?	Reject, duplicate, and reconciliation approval.
5	Incremental change, history, replay, and restart behavior is unspecified.	What detects inserts/updates/deletes, advances watermarks, and handles late data, corrections, and repeated runs?	Incremental and recovery execution.
6	Amount rules and reconciliation acceptance thresholds are unspecified.	What currency, precision, sign, rounding, comparison keys/grain, count formula, and tolerances apply?	Financial accuracy and reconciliation sign-off.
7	ETL tool, build, environment, logs, test data, and access were not identified.	Who provisions and supports the test runtime/database, files, access, reset, logs, and evidence storage?	Execution readiness.
8	No defect tracker/workflow, named owners, schedule, severity gates, or approvers were confirmed.	What tracker, cadence, dates/resources, defect thresholds, and authorized signatories should be used?	Governance, schedule, and closure.
9	Applicable security/privacy and non-functional requirements are not stated.	What data classification, masking/access/retention rules, performance targets, recovery objectives, and additional tests apply?	Data access approval and scope acceptance.

Human Review Gate

Document status: DRAFT — PENDING HUMAN REVIEW. Not approved.

Documents used: Banking Data Warehouse BRD v1.0; Banking ETL Test Strategy v1.0 (Draft — Pending Human Review); Testplan Creator guidance; supplied Example-Test-template.md, used for section order and planning style. The organization's company-details file was present but blank.

Assumptions / provisional decisions: User confirmed this is a Banking ETL project plan and authorized proceeding with the unapproved draft strategy as a provisional input. BRD identifiers BR-01–BR-10 are labels assigned here to its ten numbered requirements. No acceptance criteria, project tool, named role, environment endpoint, test date, or technical behavior was invented. Scenario priorities remain TBD.

Could not be confirmed: Strategy approval, source-to-target mappings, target schemas, detailed business rules, duplicate/reject policy, incremental/recovery design, actual files and test data, environment/tool versions, reconciliation thresholds, non-functional targets, schedule/resources, defect workflow, and named approvers.

Approval blockers: Review/approve the Test Strategy; resolve Questions 2–7 to establish expected results and safe execution readiness; define applicable data protection/non-functional scope and governance; obtain authorized approvals.

Review request: QA, business/data, and ETL/technical owners should correct this draft, answer the questions, confirm measurable entry/exit criteria and scenario priorities, and record a decision. Detailed test cases, SQL scripts, and automation are not included.

Decision: ☐ Approve after updates ☐ Return with edits Reviewer: _____ Date: _____

End of draft — do not treat as baselined or approved until the authorized human reviewers have recorded their decision.